Theft, Trade, and Social Order

Randy Hill

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The University of Arizona

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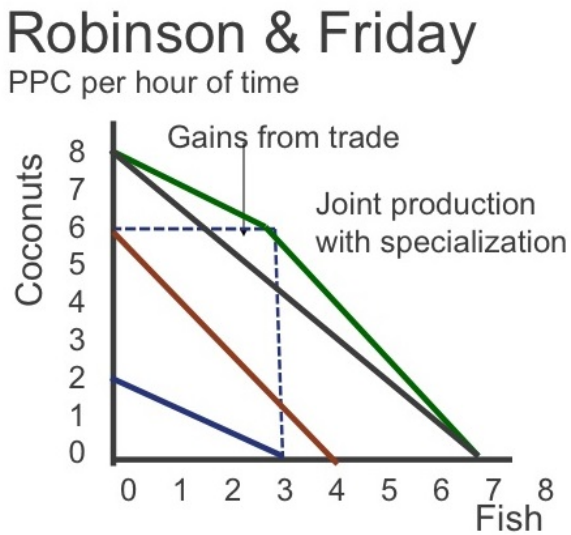
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The Fish and the Coconuts





What happens if the people on the island steal from each other?

[1] https://tamoclass.wordpress.com/

Findings

Research question: In the absence of external institutions that protect property rights, what conditions are required for peaceful exchange?

- Goods are distributed according to power
- Peaceful trade is possible only when distribution aligns with power
- Transfers can align distribution to enable peaceful trade

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Related literature

Institutions constrain violence and sustain order

Weber (1946); Olson (1971, 1993); Weingast (1995); Skaperdas & Syropoulos (1997); North, Summerhill, and Weingast (2000); North, Wallis, and Weingast (2009); Cox, North, and Weingast (2019); Wallis (forthcoming); Greif (2006); Acemoglu and Robinson (2006)

Resources allocated to conflict and consumption

Skogh and Stuart (1982); Skaperdas (1992); De Meza and Gould (1992); Hirshleifer (1995); Grossman and Kim (1996); Grossman (2002); Bates, Greif, and Singh (2002); Baker, Bulte, and Weisdorf (2010); Van Besouw, Ansink, and van Bavel (2016,2017)

Insecure property affects the gains from trade

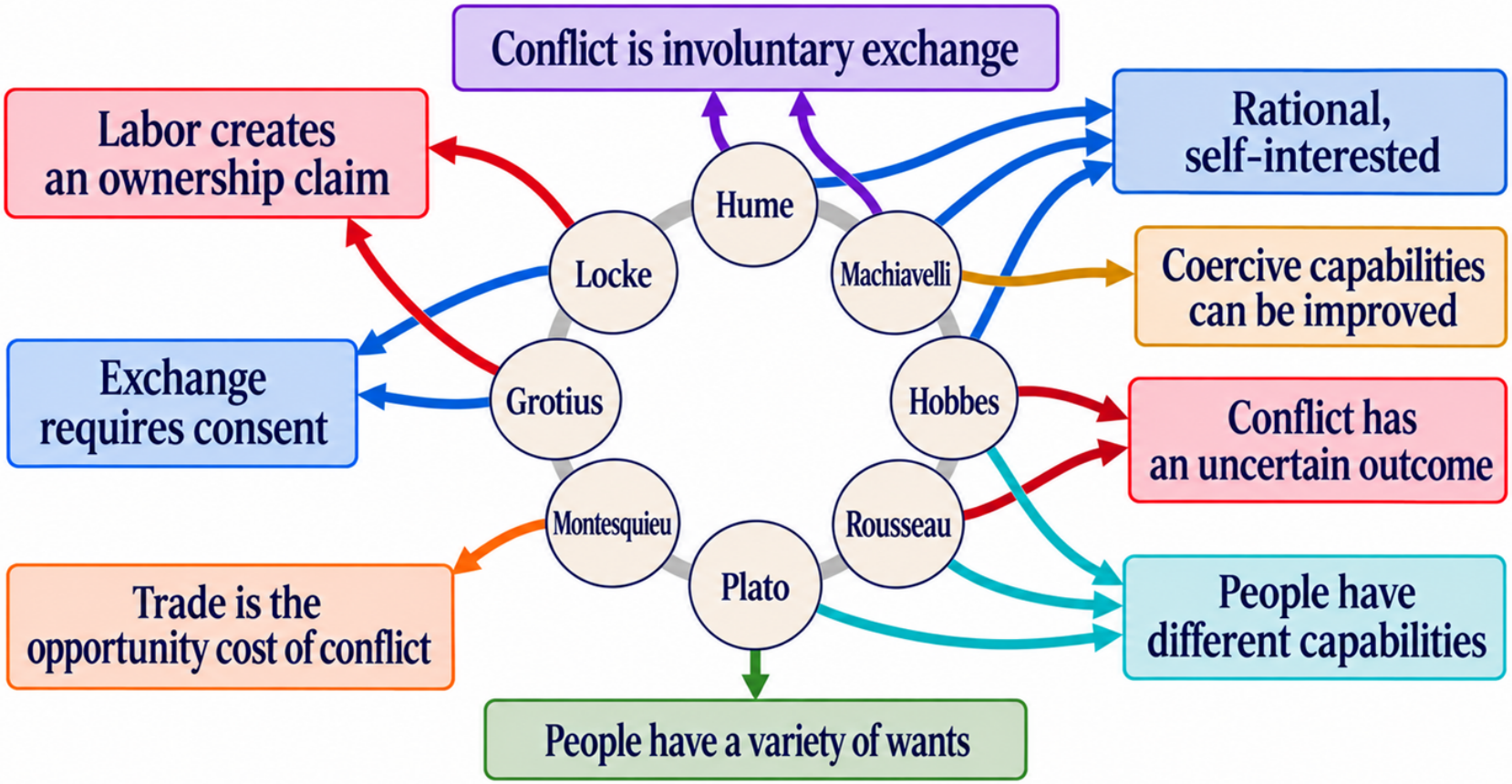
Polachek (1980); Martin, Mayer, and Thoenig (2008); Skaperdas and Syropoulos (2001, 2002)

Equilibrium in the Jungle

Piccione and Rubinstein (2007)

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Philosophical grounding



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Institutions

A Market is an institution

- Fungible and transferable goods
- Price taking
- Budget constraint binds/no credit
- Market clearing

Two-way consent for trade implies a particular form of property rights

Players accept the institution when they decide to Trade

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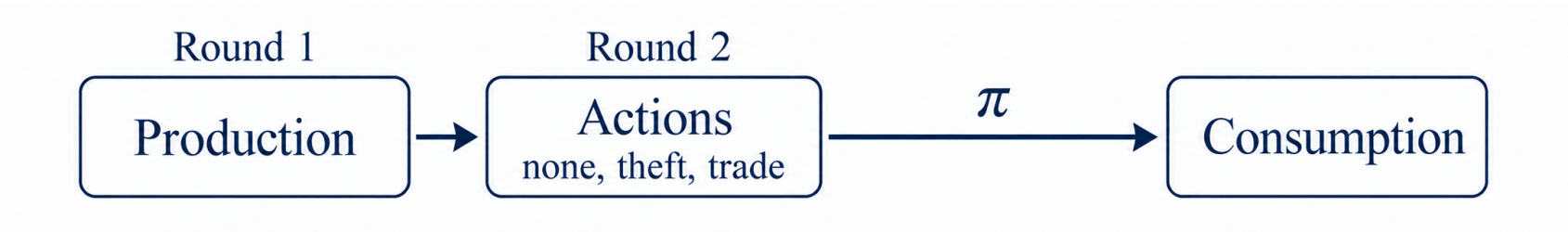
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The Theft and Trade Game

A Theft and Trade game is a two-stage dynamic game - $\Gamma = (\mathcal{N}, \Sigma, \pi)$
Finite set of 2 or more players

$$\mathcal{N} = \{1, \ldots, N\}$$

Players produce and consume goods
Players choose whether to trade with or steal from others



- Round 1 - Players allocate labor to production functions
- Round 2 - Players choose actions towards others

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Round 1 - Production

Players produce M consumer goods and one *security good*

Endowment: labor $\tau > 0$, and production technologies

$$f_{nm} : \mathbb{R}_+ \to \mathbb{R}_+, \qquad m = 1, \ldots, M, \qquad f_{nS} : \mathbb{R}_+ \to \mathbb{R}_+$$

Each f_{nm} and f_{nS} is continuous at 0, C^2, increasing, and strictly concave, with

$$f_{nm}(0) = f_{nS}(0) = 0$$

In round 1, each player simultaneously chooses a labor allocation

$$L_n = (\mathbf{t}_n, s_n) \in \mathcal{L}_n, \qquad s_n + \sum_{m=1}^{M} t_{nm} = \tau$$

s_n is player n's *security budget*
Production is completed before round 2 begins

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Production and Preferences

Player n's produced consumer-good bundle is

$$\mathbf{x}_n^P(L_n) = \left(f_{n1}(t_{n1}), \ldots, f_{nM}(t_{nM})\right),$$

and effective security (or *power*) is $S_n(L_n) = f_{nS}(s_n)$

All players have common utility

$$U : \mathbb{R}_+^M \to \mathbb{R}_+$$

$U \in C^2$ is increasing, strictly quasi-concave, and homogeneous of degree one

Every consumption good is essential and

$$U(\mathbf{0}) = 0$$

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Round 2 - Postures/Actions

In round two, each player simultaneously chooses a *posture* towards every other player.

$$a_{nj} \in \{\text{none}, \text{theft}, \text{trade}\}$$

A round two action is a vector of postures

$$a_n = (a_{nj})_{j \neq n} = \rho_n(L)$$

The action profile is

$$a = (a_1, \ldots, a_N)$$

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Information

Complete and imperfect information

- Endowments and preferences are common knowledge
- Players make simultaneous choices within each round
- Players observe labor allocations before choosing round two actions

Strategies

Player n's pure strategy is the ordered pair

$$\sigma_n = (L_n, \rho_n)$$

L_n is player n's round-one labor allocation; ρ_n is her round-two contingent posture rule, as already defined.

$$L = (L_1, \ldots, L_N), \qquad \Sigma = \left\{\sigma = (\sigma_1, \ldots, \sigma_N) : \sigma_n = (L_n, \rho_n)\ \forall\, n\right\}$$

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Partition of Players

The strategy profile σ, together with the rule that Conflict takes precedence over Trade, induces a partition of players

The payoff function π is defined over this partition

Coalitions are added by refining this partition

Payoffs

The payoff function is

$$\pi : \Sigma \to \mathbb{R}^N \qquad \pi = v \circ g$$

g is an outcome function; v is a value function

$$g : \Sigma \to \Delta\left((\mathbb{R}_+^M)^N\right) \qquad v : \Delta\left((\mathbb{R}_+^M)^N\right) \to \mathbb{R}_+^N$$

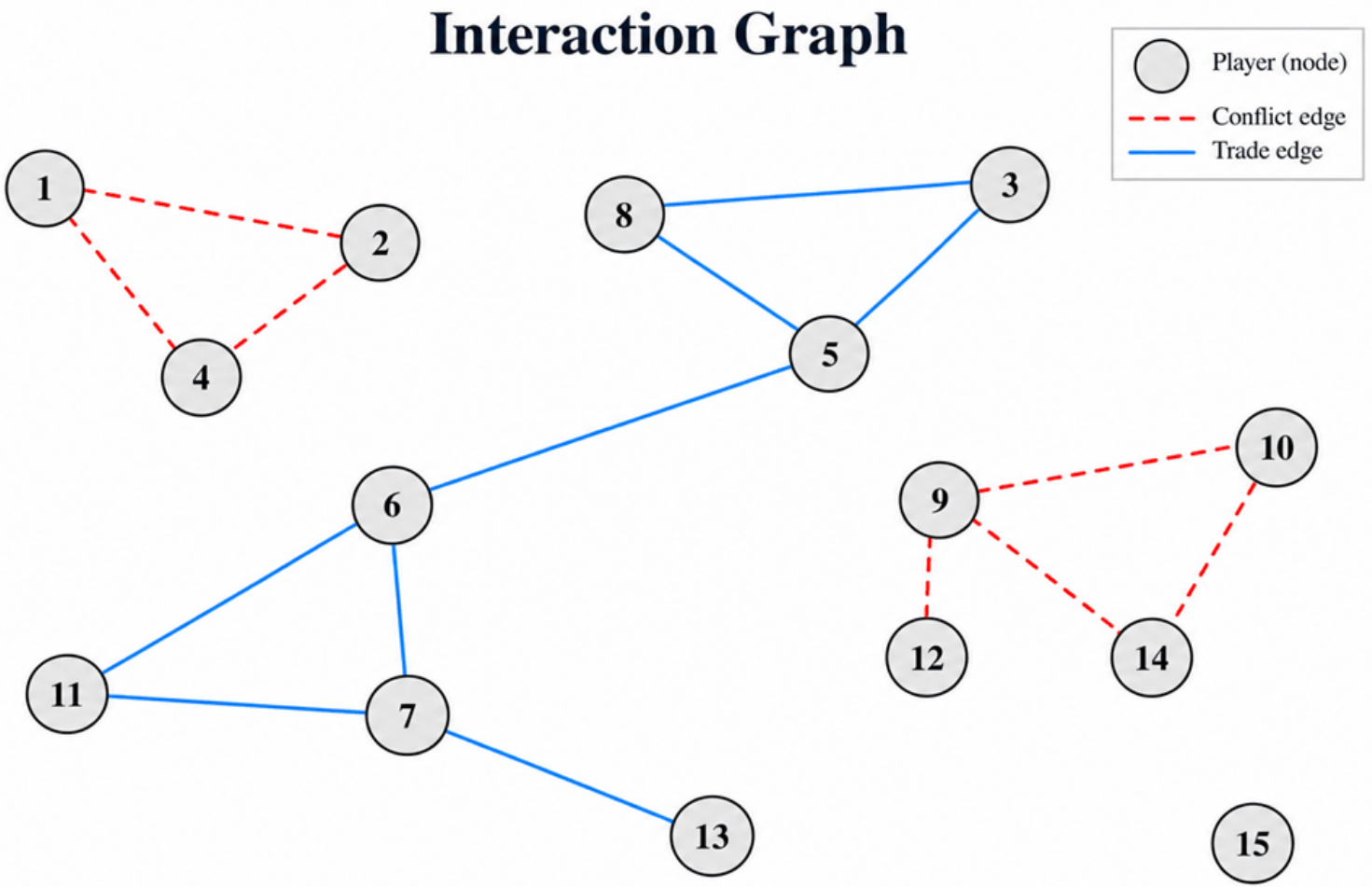
The outcome function generates a lottery over players' consumption bundles for each set in the partition of players

For each strategy profile $\sigma \in \Sigma$,

$$\pi(\sigma) = \left(\mathbb{E}_{g(\sigma)}\left[U(\mathbf{X}_1)\right], \ldots, \mathbb{E}_{g(\sigma)}\left[U(\mathbf{X}_N)\right]\right).$$



Interaction Graph



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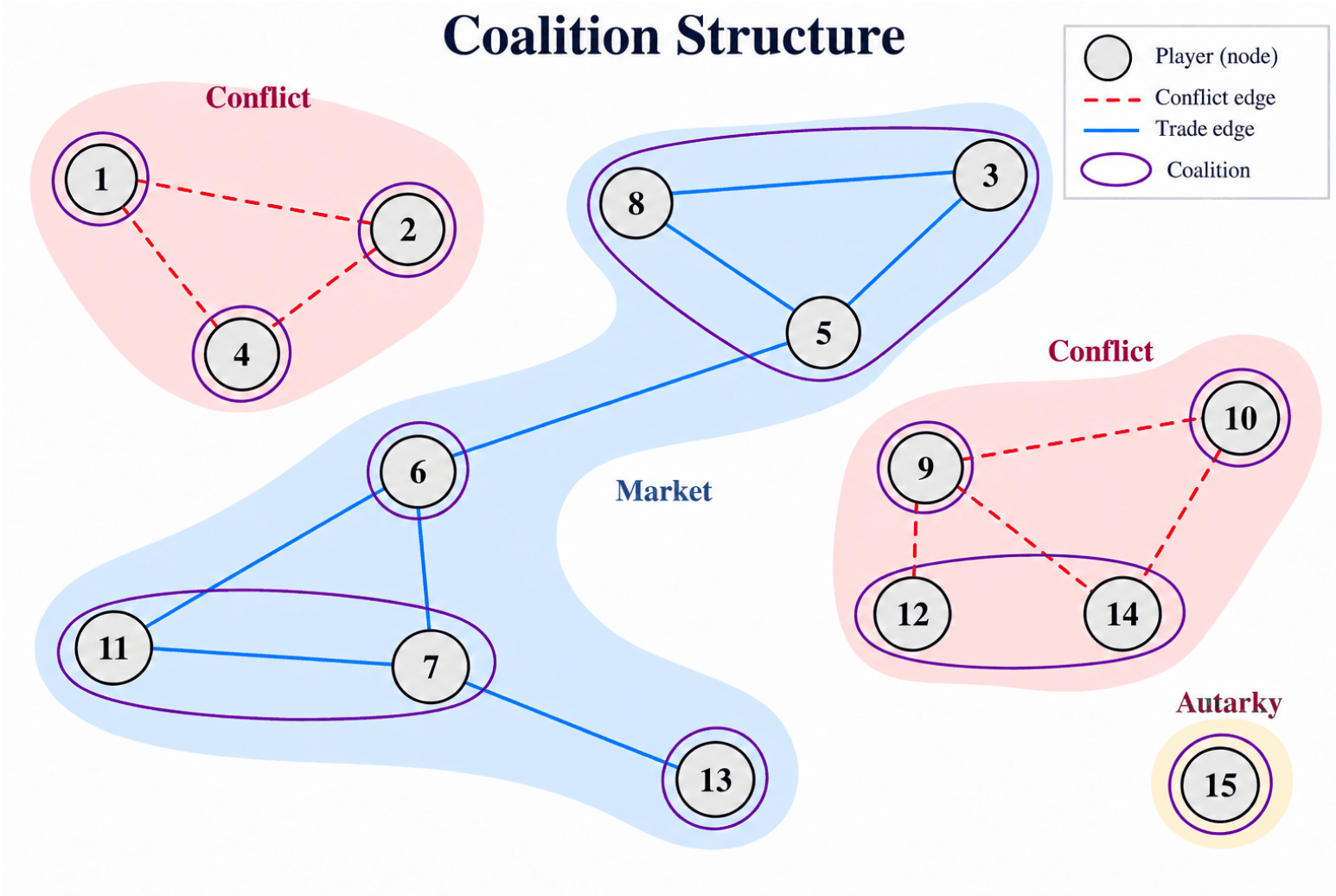
Coalition Structure

A *coalition structure* $\mathcal{B}$ refines the partition of $\mathcal{N}$ defined by the Conflict Structure and Trade Structure

A *Coalition* $B \in \mathcal{B}$ is any subset of players in the same connected component of the graph G where no pair of players share a Conflict edge.

A coalition requires some rule that distributes outcomes among its members

Full Partition on $\mathcal{N}$



Conflict Outcomes

For coalition B, coalition security is $S_B = \sum_{n \in B} S_n$

The probability that B wins in Conflict is

$$q_B = \frac{S_B}{\sum_{B' \in \mathcal{B}} S_{B'}}$$

A Conflict is winner-take-all

$$Y(H) = \sum_{n \in V(H)} \mathbf{x}_n^P$$

defines the aggregate consumer good production of a set of players

Each Conflict is an independent contest

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Trade and Autarky

All participants in a Market are price takers

A market coalition receives the competitive market allocation $\mathbf{X}_B^T$

An isolated vertex is Autarkic

$$\mathbf{X}_n^A = \mathbf{X}_n^P$$

These lotteries are degenerate

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Payoffs

An allocation is a profile of consumption bundles, one for each player

The outcome function maps each strategy profile to a lottery over allocations

$$g : \Sigma \to \Delta\left(\left(\mathbb{R}_+^M\right)^N\right)$$

The value function v assigns each player their expected utility of consumption under the lottery defined by g

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Equilibrium Concept

Subgame perfection is the natural equilibrium concept.

Social Order - an SPE where a Market exists

Trade equilibrium - an SPE where every player is in a Market

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Conflict is the Outside Option

Trade requires mutual consent:

$$\{i, j\} \in E^T \iff a_{ij} = \text{trade } \textbf{and } a_{ji} = \text{trade}$$

Conflict does not:

$$\{i, j\} \in E^C \iff a_{ij} = \text{theft } \textbf{or } a_{ji} = \text{theft}$$

Trade must be preferred by everyone.
Conflict only has to be preferred by someone.

Trade or Autarky is the old binary choice

Now there is a third option

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Existence of an SPE

Proposition: Existence of a Hobbesian Warre SPE

There exists a strategy profile $\sigma^* = (L^*, \rho^*)$ satisfying the following conditions:

(i) Every player chooses 'theft'

(ii) For every $i \in \mathcal{N}$,

$$\pi_i^C(L_i^*, L_{-i}^*) \geq \pi_i^C(L_i, L_{-i}^*) \qquad \forall\, L_i \in \mathcal{L}_i$$

Every such strategy profile is a subgame-perfect equilibrium of the theft and trade game.

War is always attainable, and any single player can force it

[1] Proof invokes Reny (1999), On the Existence of Pure and Mixed Strategy Nash Equilibria in Discontinuous Games

Characterization of SPE

Theorem: Characterizing SPE in Theft and Trade Games

In an arbitrary theft and trade game with N players and M consumer goods, any SPE $\sigma^* = (\sigma_1^*, \ldots, \sigma_N^*)$ satisfies

$$\pi_n(\sigma^*) = q_n(s^*)U(Y(\mathcal{N})) \qquad \forall\, n \in \mathcal{N},$$

where L^* is a Nash equilibrium of the universal-Conflict labor-allocation game.

Every equilibrium pays each player exactly what Warre would, whether she trades, fights, or stays home

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Structural Irrelevance

Corollary: Structural Irrelevance

For any SPE σ^* and any component H of $G(\sigma^*)$, $Y(H)$ is a nonnegative scalar multiple of $Y(N)$: every component's production is allocated in exactly the same proportions as whole-economy production.

Threat of theft, not arbitrage, enforces the law of one price

The 'law of one power price' holds across Markets, Conflicts and Autarky

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Common Production

Proposition: Common Labor Allocation

Fix a security allocation s with $0 < s_n \leq \tau$ for every $n \in \mathcal{N}$ and $s_n < \tau$ for at least one n. Consider the production games induced by a single connected Market and a single universal Conflict among all players in $\mathcal{N}$. The two games have the same unique Nash equilibrium in consumption-good labor allocations:

$$t^{T*} = t^{C*}.$$

$$\overbrace{\arg\max_{\mathbf{t}_n} \; q_n \, U\left(\sum_{i \in \mathcal{N}} \mathbf{X}_i^P(\mathbf{t}_i)\right)}^{\text{Conflict}} = \overbrace{\arg\max_{\mathbf{t}_n} \; U\left(\sum_{i \in \mathcal{N}} \mathbf{X}_i^P(\mathbf{t}_i)\right)}^{\text{Invisible hand}}$$

Invisible hand extends its reach

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Social Order and Trade

Proposition: Social Order Equilibria

A Social Order equilibrium exists if and only if there exist a Nash equilibrium L^* of the universal-Conflict labor-allocation game and a set $I \subseteq \mathcal{N}$ with $|I| \geq 2$ such that, at L^*,

(i) $Y(I) = \left(\sum_{i \in I} q_i(s^*) \right) Y(\mathcal{N})$, and

(ii) when the players of I form a Market, $\alpha_i = \dfrac{q_i(s^*)}{\sum_{j \in I} q_j(s^*)}$ for every $i \in I$.

Peace is possible, but only when markets mirror coercion

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Coalition Distribution rules

Possible Coalition Rules

Equal Shares

align group & individual optimization

but invite defection —
the grand coalition unravels

Proportional Shares

maintain individual incentives

but invite rent seeking

[1] See Nitzan (1991), Skaperdas (1996), and Münster (2009) for background.

Full Dissipation

Proposition: Nested Proportional Sharing

Suppose every $i \in B$ receives the security-weighted share S_i/S_B of coalition B's winnings, and B itself receives the security-weighted share S_B/S_H of component H. Then for every $i \in B$,

$$\pi_i(\sigma^*) = \frac{S_i}{S_B} \cdot \frac{S_B}{S_H} U\left(Y(H)\right) = \frac{S_i}{S_H} U\left(Y(H)\right),$$

exactly the payoff i would receive as a lone combatant in H. Coalition membership is payoff-irrelevant: it changes neither i's equilibrium payoff nor, since her objective is unchanged, her optimal production and security choices.

In a coalition, no one ends up better off

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Transfers

Corollary: Transfer Schedule Exists

Let $\alpha_n(\sigma)$ denote player n's payoff share under ordinary market clearing, absent any transfer. Define

$$r_n(\sigma) \equiv \left(q_n(\sigma) - \alpha_n(\sigma)\right) U\left(Y(\mathcal{N})\right).$$

If every player's post-transfer payoff is $\alpha_n(\sigma)U\left(Y(\mathcal{N})\right) + r_n(\sigma) = q_n(\sigma)U\left(Y(\mathcal{N})\right)$, then σ satisfies the necessary condition of Theorem 1 for every n.

Transfer schedule always exists to align the Market and Conflict allocations

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Institutions

- Goods are distributed according to power
- Peaceful trade is possible only when distribution aligns with power
- Transfers can align distribution to enable peaceful trade

Institutions are required to create the rules for transfers



Why peaceful exchange does not escape power



Markets	Organizations	Transfers
Coordinate exchange; they do not enforce its allocation	An organization able to restrain others possesses power itself	Describe the peaceful allocation, but cannot implement

No enforcement outside the power structure

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Conclusion

The fundamental political dilemma of an economic system is this: A government strong enough to protect property rights and enforce contracts is also strong enough to confiscate the wealth of its citizens. - Barry Weingast

Major questions:

- To what degree does coercion still play a role in the distribution of goods in the modern economy?
- Can we model institutions from primitives? Do we require fundamentally different assumptions to define institutions?

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Interaction Graph

An action profile induces the interaction graph

$$G(a) = (\mathcal{N}, E^C, E^T).$$

A Conflict edge exists if either player chooses theft:

$$\{i, j\} \in E^C \iff a_{ij} = \text{theft or } a_{ji} = \text{theft}.$$

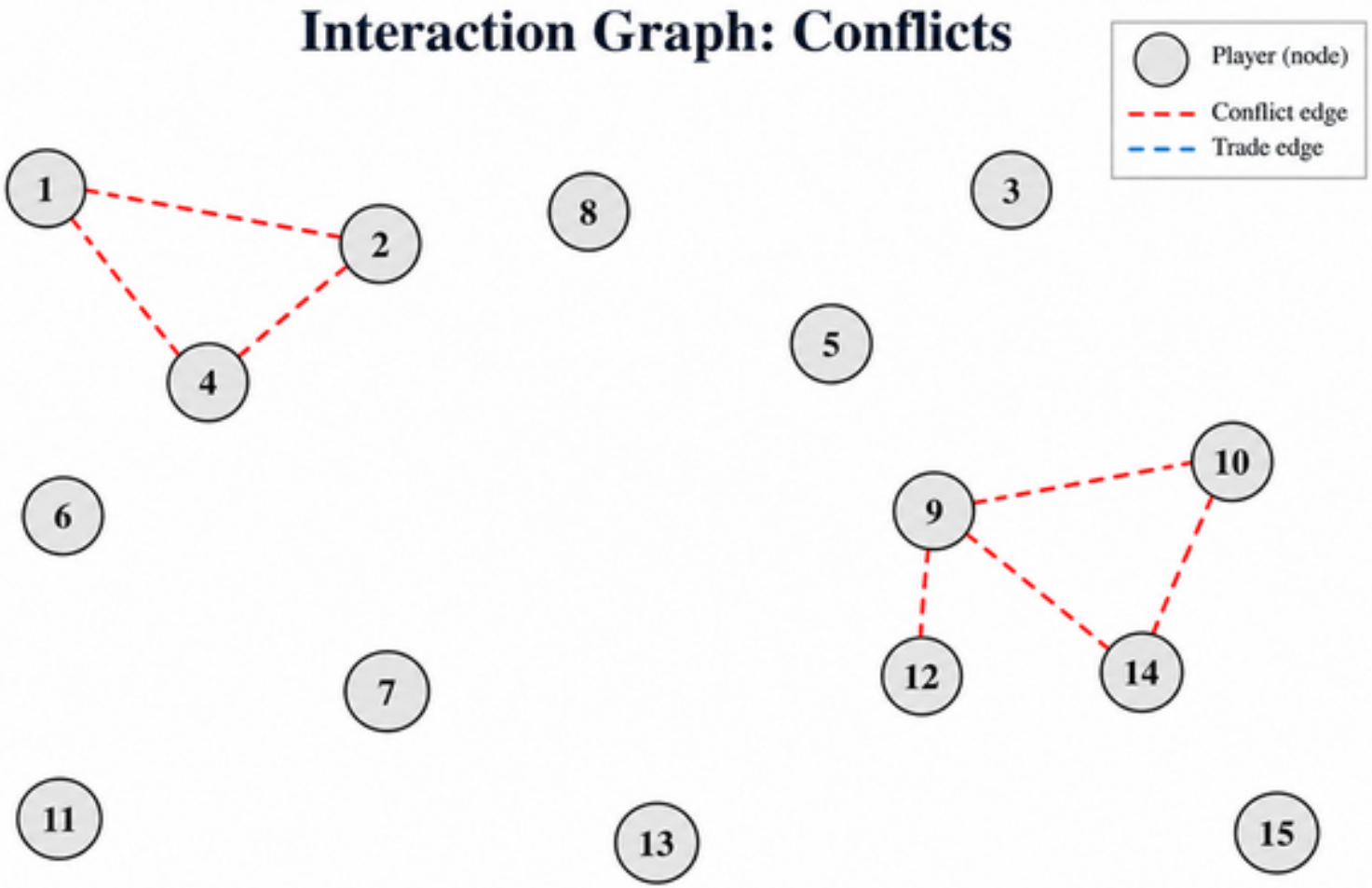
The Conflict subgraph is a *Conflict Structure*

Nontrivial connected component of the Conflict Structure is a *Conflict*

The set of players belonging to any Conflict is the set of combatants $\mathcal{N}^C$

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Conflict Structure



Trade and Autarky

Conflict takes precedence over Trade

Among noncombatants, a Trade edge exists when both players choose trade:

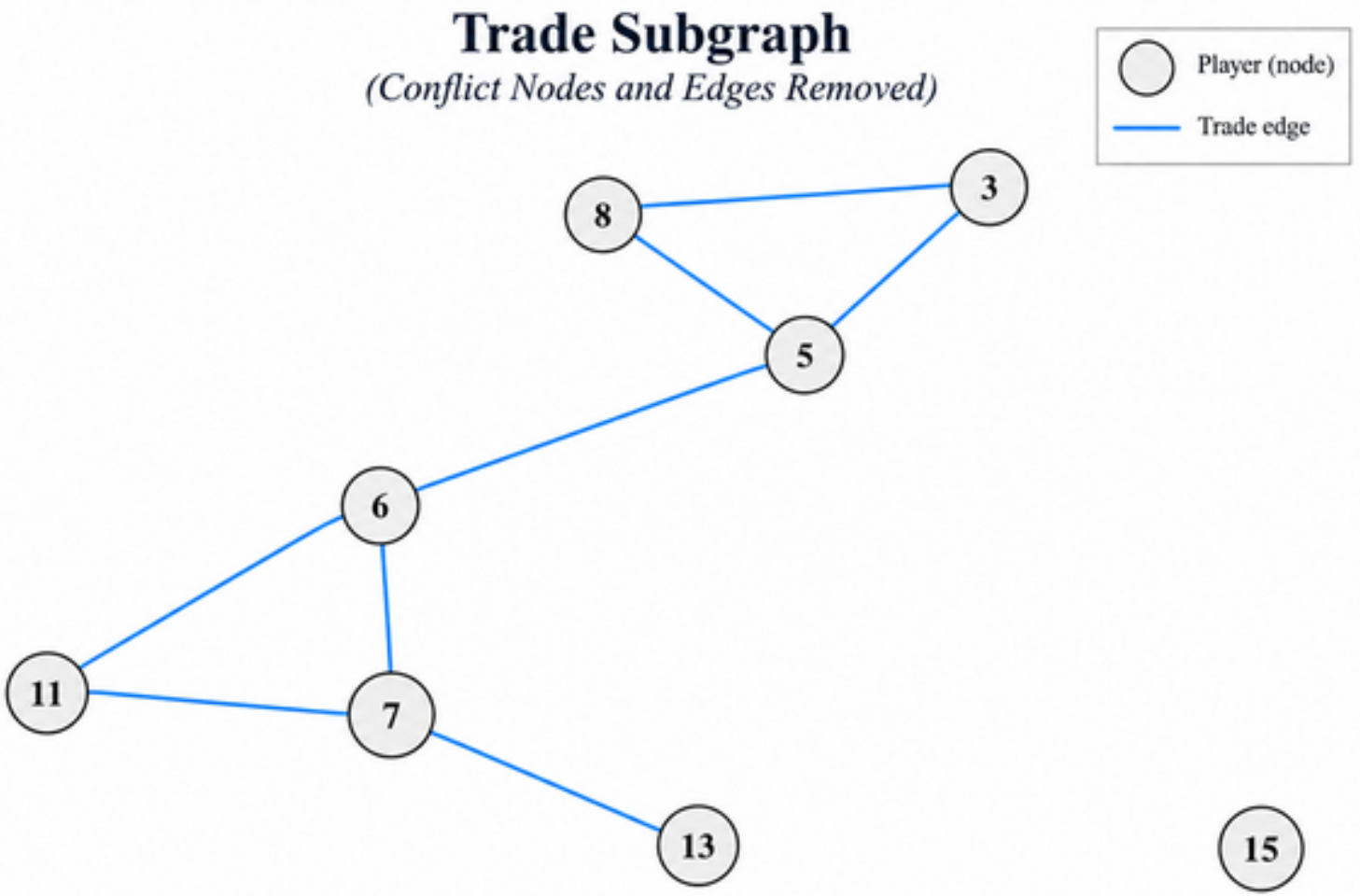
$$\{i, j\} \in E^T \iff a_{ij} = a_{ji} = \text{trade}, \qquad i, j \notin \mathcal{N}^C$$

Trade subgraph is the *Trade Structure*

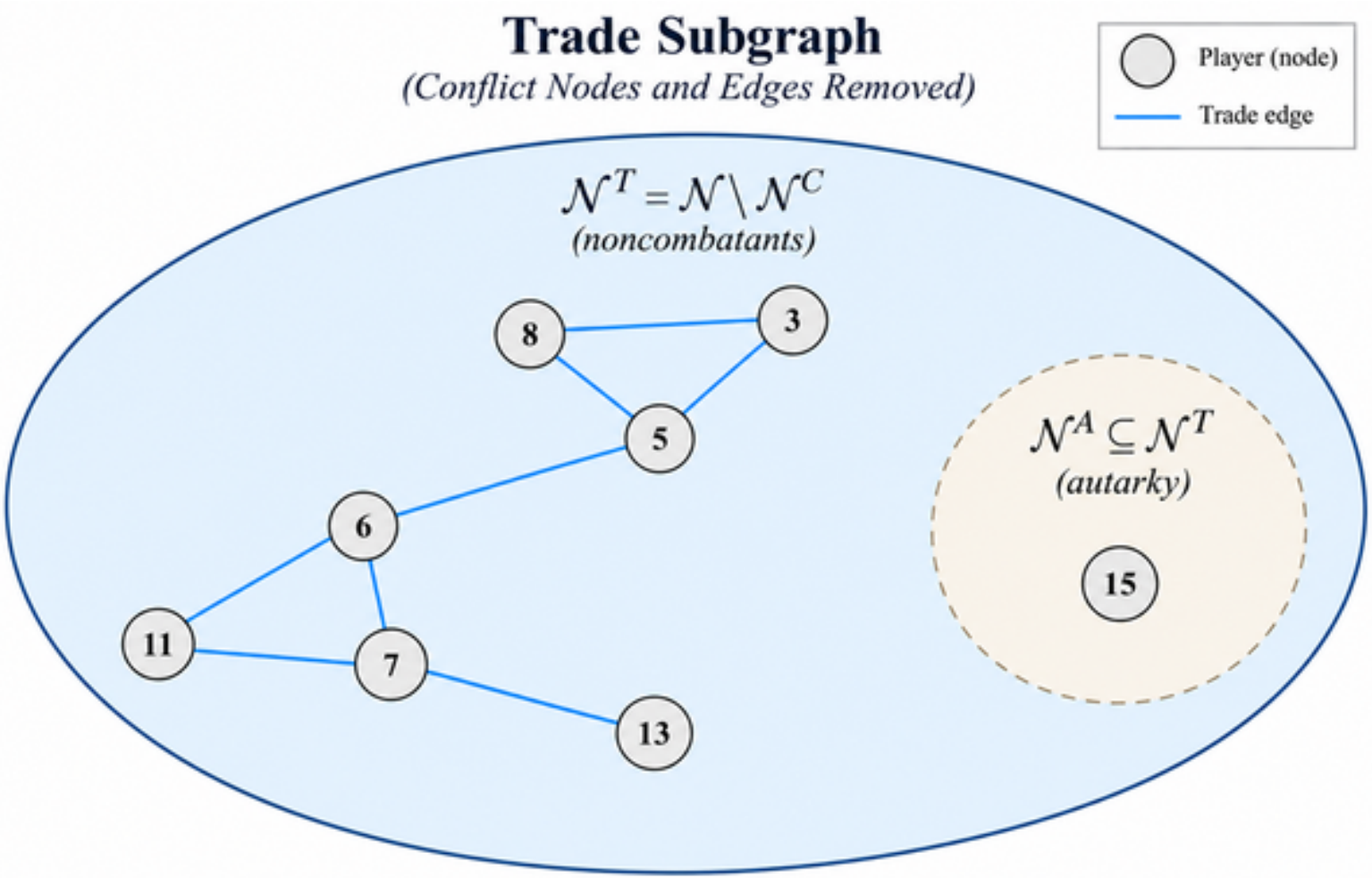
Nontrivial connected components of the Trade subgraph are *Markets*
Noncombatants who belong to no market are Autarkic.



Trade Structure



Autarky



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Additional results

- Autarky exclusion
- Security Budget
- Grand Coalition fails



Autarky

Corollary: Autarky Exclusion

If an SPE contains an Autarkic player n, then for every pair of consumption goods m, m',

$$\frac{f'_{nm}(t_{nm})}{f'_{nm'}(t_{nm'})} = \frac{\partial U(Y(\mathcal{N}))/\partial X_{m'}}{\partial U(Y(\mathcal{N}))/\partial X_m}.$$

Thus, Autarky can occur in equilibrium only if no gains from unilateral specialization remain for that player.

Autarky survives only if a player has nothing to gain from specializing

Security Budget Criterion

Lemma: Security Budget

Fix a region of $\mathcal{L}_n$ induced by σ_{-n}, with win probability $q_n(s_n)$ for the singleton coalition $\{n\}$, where $S_{-n} \equiv \sum_{j \neq n} S_j$ is fixed. Player n's optimal security budget s_n satisfies, for every consumption good m,

$$1 - q_n(s_n) = \frac{\frac{d}{dt_{nm}} \ln U\left(Y(H)\right)}{\frac{d}{ds_n} \ln f_{nS}(s_n)},$$

and this equation has a unique solution.

Every player picks the security budget to balance costs and benefits

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Grand Coalition

Theorem: Grand Coalition Instability

The grand coalition $B = \mathcal{N}$ splitting output equally cannot be an SPE.

With no benefit tied to security, joint surplus $U(Y(\mathcal{N}))$ is strictly decreasing in every s_n, so the surplus-maximizing choice is $s_n = 0$ for every n

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